

EXPERIMENT – 2

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Section: 8

Title

File Copying Using Low-Level System Calls and System Call Tracing using strace

Aim

To implement a file-copy program using the low-level system calls `open()`, `read()`, `write()`, and `close()`, and trace the `cat` command using `strace`.

Theory

The `open()` system call is used to open a file, `read()` is used to read data from a file, `write()` is used to write data to a file, and `close()` is used to close an opened file. The `strace` command is used to trace the system calls made by a Linux command.

Algorithm

1. Start the program.
2. Open the source file using `open()`.
3. Open or create the destination file using `open()`.
4. Read data from the source file using `read()`.
5. Write the data into the destination file using `write()`.
6. Repeat the read and write operations until the entire file is copied.
7. Close both files using `close()`.
8. Display the successful file-copy message.
9. Trace the `cat` command using `strace`.
10. Stop the program.

Code

```
k_anusri@DESKTOP-TM20BAQ:~$  
k_anusri@DESKTOP-TM20BAQ:~$ cat > exp2_filecopy.c <<'EOF'  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
#include <stdlib.h>  
  
int main()  
{  
    int source, destination;  
    char buffer[1024];  
    ssize_t n;  
  
    source = open("input.txt", O_RDONLY);  
  
    if (source < 0)  
    {  
        perror("open");  
        return 1;  
    }  
  
    destination = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    if (destination < 0)  
    {  
        perror("open");  
        close(source);  
        return 1;  
    }  
  
    while ((n = read(source, buffer, sizeof(buffer))) > 0)  
    {  
        write(destination, buffer, n);  
    }  
  
    close(source);  
    close(destination);  
  
    printf("File copied successfully.\n");  
  
    return 0;  
}
```

Execution Screenshots

Screenshot 1 — Create input file

```
k_anusri@DESKTOP-TM20BAQ:~$ echo "This is Experiment 2 file copying using Linux system calls." > input.txt
```

Screenshot 2 — Compilation and execution

```
k_anusri@DESKTOP-TM20BAQ:~$ gcc exp2_filecopy.c -o exp2_filecopy && ./exp2_filecopy  
File copied successfully.
```

Screenshot 3 — Verify copied file

```
k_anusri@DESKTOP-TM20BAQ:~$ cat output.txt  
This is Experiment 2 file copying using Linux system calls.
```

Screenshot 4 — strace

[illegible]

